

[BUG] Loading and saving models and weights with TensorFlow doesn't b

<https://github.com/PennyLaneAI/pennylane/issues/4856>

ROW 114

[BUG] Loading and saving models and weights with TensorFlow doesn't behave as expected #4856

New issue



Closed



isaacdevlugt opened on Nov 17, 2023 · edited by isaacdevlugt

Edits ...

Expected behavior

When `save_weights` and `load_weights` are called, I expect that the model's weights are the same before and after they're saved and loaded. Also, if I use `model.save` to save the whole model and `tf.keras.models.load_model`, I expect that the weights are the same.

Actual behavior

In both cases (saving just the weights or saving the whole model), the models that are created / loaded after saving give the same outputs / predictions, but when their weights are inspected, what's shown isn't the same.

Additional information

<https://discuss.pennylane.ai/t/error-reloading-circuit-from-qasm-string/3679>

Source code

```
import tensorflow as tf
import pennylane as qml
from pennylane import numpy as np

dev = qml.device("default.qubit")
n_qubits = 2

def create_model():
    @qml.qnode(dev)
    def circuit(inputs, weights):
        qml.AngleEmbedding(inputs, wires=range(n_qubits))
        qml.RX(weights[0], 0)
        qml.RX(weights[1], 1)
        return [qml.expval(qml.PauliZ(0)), qml.expval(qml.PauliZ(1))]

    weight_shapes = {"weights": (n_qubits,)}
    quantum_layer = qml.qnn.KerasLayer(circuit, weight_shapes, output_dim=2)

    model = tf.keras.models.Sequential()
    model.add(quantum_layer)
    model.add(tf.keras.layers.Dense(2, activation=tf.nn.softmax, input_shape=(2,)))

    model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
    return model

model = create_model()

num_points = 5
dummy_input_data = np.random.uniform(0, np.pi, size=(num_points, 2))
dummy_output_data = np.random.randint(2, size=(num_points, 2))

model.fit(dummy_input_data, dummy_output_data, epochs=1, batch_size=0)

model.save("model")

loaded_model = tf.keras.models.load_model("model")

print("saved weights:", model.weights)
print("Loaded weights:", loaded_model.weights)

print(model.predict(dummy_input_data) == loaded_model.predict(dummy_input_data))

saved_weights = model.weights
model.save_weights("saved_weights")

new_model = create_model()
new_model.load_weights("saved_weights")
print("Weights saved:", saved_weights)
print()
print("Loaded weights:", new_model.weights)
print()
print(model.predict(dummy_input_data) == new_model.predict(dummy_input_data))
```

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



Tracebacks

```
saved weights: [<tf.Variable 'weights:0' shape=(2,) dtype=float32, numpy=array([-0.6488751, 1.1488918],
array([[ 0.1556892, -0.646571 ],
       [-0.2932235, 1.1476017]]], dtype=float32)>, <tf.Variable 'dense/bias:0' shape=(2,) dtype=float32, numpy=
Loaded weights: [<tf.Variable 'dense/kernel:0' shape=(2, 2) dtype=float32, numpy=
array([[ 0.1556892, -0.646571 ],
       [-0.2932235, 1.1476017]]], dtype=float32)>, <tf.Variable 'dense/bias:0' shape=(2,) dtype=float32, numpy=
1/1 [=====] - 0s 30ms/step
2023-11-17 16:19:22.491641: I tensorflow/core/common_runtime/executor.cc:1197] [/device:CPU:0] (DEBUG INFO) Ex
[[{{node inputs}}]]
2023-11-17 16:19:22.502508: I tensorflow/core/common_runtime/executor.cc:1197] [/device:CPU:0] (DEBUG INFO) Ex
[[{{node Placeholder}}]]
1/1 [=====] - 0s 138ms/step
[[ True  True]
 [ True  True]
 [ True  True]
 [ True  True]
 [ True  True]]

Weights saved: [<tf.Variable 'weights:0' shape=(2,) dtype=float32, numpy=array([-0.6488751, 1.1488918], dtype=
array([[ 0.1556892, -0.646571 ],
       [-0.2932235, 1.1476017]]], dtype=float32)>, <tf.Variable 'dense/bias:0' shape=(2,) dtype=float32, numpy=
Loaded weights: [<tf.Variable 'weights:0' shape=(2,) dtype=float32, numpy=array([-0.6488751, 1.1488918], dtype=
1/1 [=====] - 0s 31ms/step
1/1 [=====] - 0s 39ms/step
[[ True  True]
 [ True  True]
 [ True  True]
 [ True  True]
 [ True  True]]
```

System information

```
Name: PennyLane
Version: 0.33.1
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/PennyLaneAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /Users/isaac/.virtualenvs/pennylane-tensorflow/lib/python3.9/site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, s
Required-by: PennyLane-Lightning, PennyLane-qiskit

Platform info:      macOS-14.1.1-x86_64-i386-64bit
Python version:     3.9.14
Numpy version:      1.23.5
Scipy version:      1.11.3
Installed devices:
- default.gaussian (PennyLane-0.33.1)
- default.mixed (PennyLane-0.33.1)
- default.qubit (PennyLane-0.33.1)
- default.qubit.autograd (PennyLane-0.33.1)
- default.qubit.jax (PennyLane-0.33.1)
- default.qubit.legacy (PennyLane-0.33.1)
- default.qubit.tf (PennyLane-0.33.1)
- default.qubit.torch (PennyLane-0.33.1)
- default.qutrit (PennyLane-0.33.1)
...
- qiskit.ibmq.circuit_runner (PennyLane-qiskit-0.33.0)
- qiskit.ibmq.sampler (PennyLane-qiskit-0.33.0)
- qiskit.remote (PennyLane-qiskit-0.33.0)
- lightning.qubit (PennyLane-Lightning-0.33.1)
```

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



2

 isaacdevlugt added **bug** 🐛 on Nov 17, 2023



tbromley on Nov 17, 2023

Contributor ...

Thanks @isaacdevlugt! This is interesting, because we do test for model weight saving here:
https://github.com/PennyLaneAI/pennylane/blob/master/tests/qnn/test_keras.py#L578

However, your example above places a classical layer *after* the quantum one, while our test just has a classical layer before.



lillian542 on Dec 12, 2023

Contributor ...

The user was able to solve the problem by updating how AngleEmbedding was initialized in their code, so we are closing this issue. The `model.save` function uses autograph, which implicitly adds a batch dimension.



lillian542 closed this as **completed** on Dec 12, 2023



HeyPhiS mentioned this on Jan 5, 2024

🔗 [BUG] Loading hybrid quantum neuronal networks does not work with TensorFlow #4999

[BUG] Loading hybrid quantum neuronal networks does not work with TensorFlow #4999

New issue

Closed



HeyPhiS opened on Jan 5, 2024

Expected behavior

When using the tensorflow function `model.save` und `tf.keras.models.load_model` it is expected that the model load all layers and its ways and recreates the saved model to include all weights etc.

Actual behavior

When using `model.save` and `tf.keras.models.load_model` with a hybrid model including both classical layers and `qml.qnn.KerasLayer` the network including the quantum layer is reloaded without error, however the weights of the quantum layer are not.

This issue exists independent of the order of quantum and classical layers.

Additional information

This issue was discussed in the pennylane support thread <https://discuss.pennylane.ai/t/error-reloading-circuit-from-qasm-string/3679>
This thread already gave rise to issue [#4856](#) to which this issue is possibly related

Source code

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import numpy as np

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    weight_shapes = {"weights": (n_qubits,)}
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    model = tf.keras.models.Sequential()
    model.add(quantum_layer)
    model.add(tf.keras.layers.Dense(2, activation=tf.nn.softmax, input_shape=(2,)))

    model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
    return model

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num_points = 5
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dummy_output_data = np.random.randint(2, size=(num_points, 2))

model.fit(dummy_input_data, dummy_output_data, epochs=1, batch_size=0)

model.save("model")

loaded_model = tf.keras.models.load_model("model")

print("")
print("saved weights:", model.layers[0].weights)
print("")
print("Loaded weights:", loaded_model.layers[0].weights)
```

Tracebacks

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



fix keraslayer set and get attr dunder #5008

<> Code

Merged timmysilv merged 3 commits into `v0.34.0-rc0` from `keras-attr-fix` on Jan 5, 2024

Conversation 2 Commits 3 Checks 33 Files changed 3

+44 -5



timmysilv commented on Jan 5, 2024 · edited

Contributor

Context:

Fixes #4999. We re-implemented the default `setattr` and `getattr` dunder for `KerasLayer` (needed to modify the `QNode` saved on the layer) but we should have called `super()`, because `keras.layer.Layer` has its own custom implementation that we needed to use. Our `TorchLayer` already calls `super` in these cases.

Note - our existing tests were passing because the weights used had custom names ("w1", "w2", etc.), and this issue only comes up when your weights are named "weights" (which is a rather common case, I'm sure).

Description of the Change:

- Replace our default-handling of `get/set-attr` dunder with a call to `super()`
- add a test case matching the one in the bug report using the "weights" identifier

Benefits:

Models using `KerasLayer` can be saved and loaded properly!

Possible Drawbacks:

N/A

Related GitHub Issues:

Reported initially in #4999, same as #4611 but for Keras



fix keraslayer set and get attr dunder

9dd3deb

timmysilv added this to the v0.34 milestone on Jan 5, 2024



github-actions bot commented on Jan 5, 2024

Contributor

Hello. You may have forgotten to update the changelog!

Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



Reviewers

lillian542



albi3ro



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

v0.34

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

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3 participants

